



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Advanced)
ALLEN COMMON TEST
31-10-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6
	A.	A,B,C,D	A,C,D	A,D	A,C	A,B	A,C
SECTION-II	Q.	1	2	3	4	5	6
	A.	2.33	4.44	0.66	2.25	5.00	0.00
SECTION-III	Q.	1	2	3	4	5	6
	A.	3	2	4	2	5	9

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6
	A.	A,B,D	B,C	D	A,D	A,B,D	B,C,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	0.20	4.00	169.00	1.26 to 1.27	18.00	1171.50
SECTION-III	Q.	1	2	3	4	5	6
	A.	4	3	5	4	2	2

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6
	A.	A,B	B,D	A,B,C	A,B,C	B,C,D	A,B,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	2.00	1.60	1.00	1.00	0.50	2.50
SECTION-III	Q.	1	2	3	4	5	6
	A.	8	2	2	3	7	4

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

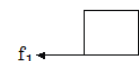
1. Ans (A,B,C,D)

$$f_{01} = 50 \text{ N}; \quad f_{02} = 100 \text{ N}$$

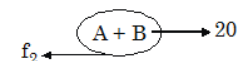
$$f_{03} = 150 \text{ N}; \quad f_{04} = 0 \text{ N}$$

If the system moves with no relative motion,
then common acceleration = 2m/s^2

$$A : f_1 = 20 \text{ N}$$

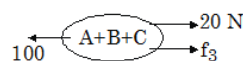


$$f_2 - 20 = 20 \times 2$$



$$f_2 = 60 \text{ N}$$

$$100 - 20 - f_3 = 30 \times 2$$



$$f_3 = 20 \text{ N}$$

2. Ans (A,C,D)

$$Z = \sqrt{20^2 + 20^2} = 20\sqrt{2} \Omega$$

$$C_0 = \frac{V_0}{Z} = 5$$

$$\tan \phi = \frac{X}{R} = \frac{20}{20} = 1; \quad \phi = 45^\circ$$

$$i(t) = 5 \sin \left(\omega t - \frac{\pi}{4} \right)$$

$$V_L(t) = 150 \sin \left(\omega t + \frac{\pi}{4} \right)$$

$$V_C(t) = 50 \sin \left(\omega t - \frac{3\pi}{4} \right)$$

3. **Ans (A,D)**

To calculate the mutual inductance of the pair of loops, let us assume that there is a current I in the bigger loop. Field due to QP and SR will be very small at the location of loop A due to large distance. Loop A is very close to arm PR and QS. We can consider PR and SQ to be of infinite length for writing magnetic field at location of A.

$$B_{PR} = \frac{\mu_0 I}{2\pi a} \odot \text{ and } B_{QS} = \frac{\mu_0 I}{2\pi(2a)} \otimes$$

$$B = \frac{\mu_0 I}{4\pi a} \odot$$

Since loop A is small, we can consider the field to be uniform throughout its area.

$$\phi_A = x^2 \cdot B = \frac{\mu_0 I}{4\pi a} x^2$$

Comparing with $\phi = MI$

$$\text{Mutual inductance of coils } M = \frac{\mu_0 x^2}{4\pi a}$$

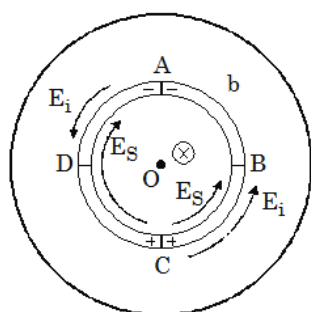
M does not change if loop A is made primary.

$\therefore$ emf induced in B when current is made to change in A is

$$\varepsilon_B = M \frac{di_A}{dt} = \frac{\mu_0 x^2 \alpha}{4\pi a}$$

4. **Ans (A,C)**

E_s = electrostatic field due to accumulated charge.



E_i = induced $\vec{E}$ due to varying $\vec{B}$

$$(E_i + E_s) = E_1$$

$$(E_i - E_s) = E_2$$

5. **Ans (A,B)**

Suppose at any time t_1 rod MN is at a distance x from the resistance R . Velocity of rod at that instant is

$$v = \frac{dx}{dt}$$

$$\text{Motional emf } e = Bv\ell \quad \dots(1)$$

Total resistance R_{net} of the circuit at this instant is

$$R_{\text{net}} = R + 2\lambda x$$

$$\therefore i = \frac{e}{R_{\text{net}}} = \frac{B\ell v}{R + 2\lambda x} \quad (\text{anticlockwise})$$

$$\text{or } v = \frac{i(R + 2\lambda x)}{B\ell}$$

Instantaneous acceleration of the rod, is

$$a = \frac{dv}{dt} = \frac{2\lambda i}{B\ell} \cdot \frac{dx}{dt} = \frac{2\lambda i}{B\ell} \cdot v$$

$$\text{or } a = \frac{2\lambda i^2}{B^2 \ell^2} (R + 2\lambda x) \quad \dots(2)$$

At this moment two forces are acting on the rod, applied force F (towards right) and magnetic force $F_m = i\ell B$ (towards left). So

$$F_{\text{net}} = F - F_m$$

$$\text{or } ma = F - i\ell B$$

$$\therefore F = i\ell B + \frac{2m\lambda i^2}{B^2 \ell^2} (R + 2\lambda x)$$

(b) Work done by F per unit time is given by

$$P_1 = Fv = \frac{F_i(R + 2\lambda x)}{B\ell} \quad \dots(3)$$

and heat generated in circuit per unit time is

$$P_2 = i^2 R_{\text{net}} = i^2 (R + 2\lambda x) \quad \dots(4)$$

Therefore, fraction of work converted into heat is

$$f = \frac{P_2}{P_1} = \frac{iB\ell}{F}$$

Substituting value of F we get

$$f = \left\{ 1 + \frac{2m\lambda i (R + 2\lambda x)}{B^3 \ell^3} \right\}^{-1}$$

6. **Ans (A,C)**

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} - \frac{1}{12} = \frac{1}{-6}$$

$$\frac{1}{v} - \frac{1}{12} - \frac{1}{6} = \frac{-3}{12}$$

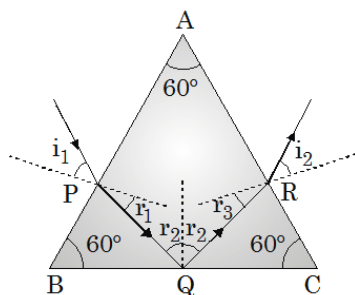
$$v = -4 \text{ cm}$$

The image formed by L_1 is virtual.

SECTION-II

1. Ans (2.33)

The ray diagram is drawn for the sake of convenience.



$$r_1 = r_3 = r$$

$$i_1 = i_2 = i$$

$$r_2 \approx C$$

$$\text{Since, } r_1 + r_2 = r_2 + r_3 = 60^\circ$$

$$\Rightarrow r_1 = r_3 = r \text{ (say)}$$

Similarly by symmetry, we have $i_1 = i_2 = i$ (say)

$$\text{Also, } r_2 \approx C$$

$$\Rightarrow r = 60^\circ - C$$

$$\text{Given, } \delta_{\text{total}} = 120^\circ$$

$$\Rightarrow \delta_P + \delta_Q + \delta_R = 120^\circ$$

$$\Rightarrow (i - r) + (180 - 2C) + (i - r) = 120^\circ$$

$$\Rightarrow 2i - 2(60^\circ - C) + 180^\circ - 2C = 120^\circ$$

$$\Rightarrow 2i = 60^\circ$$

$$\Rightarrow i = 30^\circ$$

Now, according to Snell's Law, we have

$$\mu = \frac{\sin i_1}{\sin r_1} = \frac{\sin i}{\sin r}$$

$$\text{But } r_1 + r_2 = 60^\circ$$

$$\Rightarrow r + C = 60^\circ \Rightarrow r = 60^\circ - C$$

$$\Rightarrow \mu = \frac{\sin(30^\circ)}{\sin(60^\circ - C)} \Rightarrow \mu = \frac{0.5}{\frac{\sqrt{3}}{2} \cos C - \frac{1}{2} \sin C}$$

$$\text{But } \sin C = \frac{1}{\mu}$$

$$\Rightarrow \mu = \frac{0.5}{0.5 \left(\sqrt{3} \sqrt{1 - \frac{1}{\mu^2}} - \frac{1}{\mu} \right)} \Rightarrow \mu = \frac{\mu}{\sqrt{3} \sqrt{\mu^2 - 1} - 1}$$

$$\Rightarrow \sqrt{3} \sqrt{\mu^2 - 1} = 2 \Rightarrow 3(\mu^2 - 1) = 4$$

$$\Rightarrow 3\mu^2 = 7 \Rightarrow \mu = \sqrt{\frac{7}{3}} \Rightarrow \mu = 1.52$$

2. Ans (4.44)

$$P_0 = 2P_{L_1} + 2P_2 + P_M$$

$$P_{L_1} = \frac{1}{f_{L_1}} = (1.5 - 1) \left(\frac{1}{20} - \frac{1}{-20} \right) = \frac{1}{20}$$

$$P_{L_2} = \frac{1}{f_{L_2}} = (2 - 1) \left(\frac{1}{-20} - \frac{1}{20} \right) = -\frac{1}{10}$$

$$P_M = -\frac{1}{f_M} = -\frac{1}{R/2} = -\frac{1}{10}$$

$$P_0 = 2 \left(\frac{1}{20} \right) + 2 \left(-\frac{1}{10} \right) - \frac{1}{10}$$

$$P_0 = -\frac{1}{5\text{cm}} = \left[\frac{-1}{0.05\text{m}} \right] = -20 \text{ diopter}$$

$$f_0 = \frac{1}{-P_0} = (5\text{cm})$$

$$f_0 = 5 \text{ cm}$$

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f_0}$$

$$\frac{1}{v} + \frac{1}{-40} = \frac{1}{5}$$

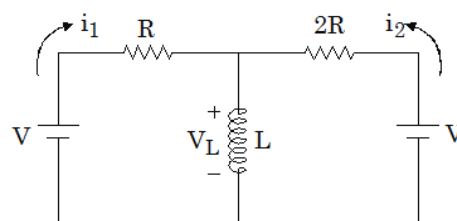
$$\frac{1}{v} = \frac{1}{40} + \frac{1}{5} = \frac{1+8}{40}$$

$$v = \frac{40}{9} \text{ cm}$$

3. Ans (0.66)

$$i_0 = \left(\frac{V}{2R} \right)$$

Just after key K_1 is closed,



$$i_1 + i_2 = i_0 = \frac{V}{2R} \quad \dots(1)$$

$$V - i_1 R = V_L \quad \dots(2)$$

$$V - i_2 (2R) = V_L \quad \dots(3)$$

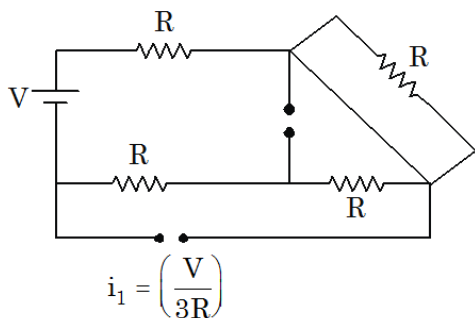
$$\text{Solving } V_L = \frac{2}{3} V$$

$$\frac{2V}{3} = \frac{L di}{dt}$$

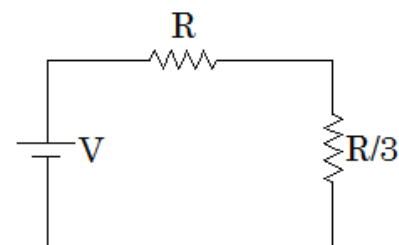
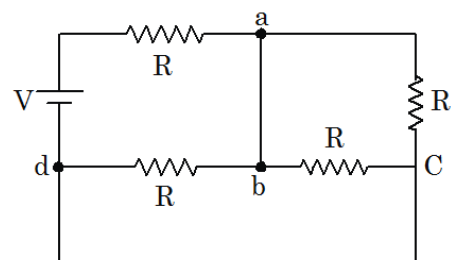
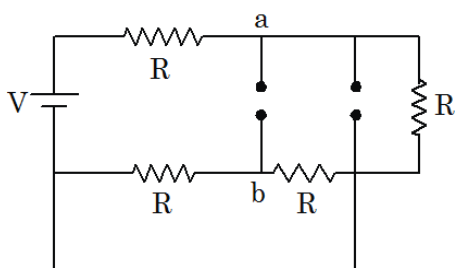
$$\frac{di}{dt} = \frac{2}{3} \left(\frac{V}{L} \right)$$

4. Ans (2.25)

$t = 0$



$t = \infty$



$$i_2 = \frac{3V}{4R}$$

$$\frac{i_1}{i_2} = \frac{V}{3R} \times \frac{4R}{3V} = \left(\frac{4}{9} \right)$$

$$\frac{i_2}{i_1} = \left(\frac{9}{4} \right)$$

5. Ans (5.00)

Here $m = 30$, $f_e = 5$ cm, $D = 25$ cm

Magnifying power of a compound microscope is

$$m = m_o \times m_e = m_o \left(1 + \frac{D}{f_e} \right)$$

$$\text{or } 30 = m_o \left(1 + \frac{25}{5} \right) \quad \text{or } m_o = 5$$

6. Ans (0.00)

$$\frac{1}{v} + \frac{1}{-10} = \frac{1}{10}$$

$$\frac{1}{v} = \frac{1}{5}; v = 5 \text{ cm}$$

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$-\frac{1}{v^2} \left(\frac{dv}{dt} \right) - \frac{1}{u^2} \left(\frac{du}{dt} \right) = 0$$

$$\vec{v}_1 = -\frac{v^2}{u^2} \cdot \vec{v}_0$$

$$v_{I,M} = -2\hat{i} \text{ m/s}$$

$$\vec{v}_{I,g} = \vec{v}_{I,M} + \vec{v}_{M,g} = 0$$

SECTION-III

1. Ans (3)

$$\Delta x = t_1 \left(1 - \frac{1}{\mu_1/\mu_0} \right) + t_2 \left(1 - \frac{1}{\mu_2/\mu_0} \right) + t_3 \left(1 - \frac{1}{\mu_3/\mu_0} \right)$$

$$= 3 \text{ cm}$$

The image formed by slabs is lying at centre of curvature of mirror.

2. Ans (2)

$$L_1 = \frac{\mu_0 N_1^2 A}{l}, L_2 = \frac{\mu_0 N_2^2 A \mu_r}{l}$$

Since $V = L \frac{dl}{dt}$, So

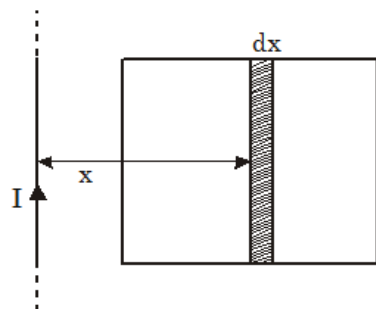
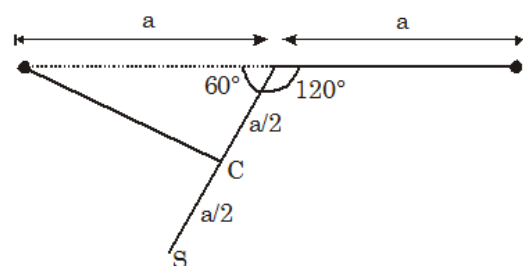
$$\frac{V_1}{V_2} = \frac{L_1}{L_2} = \left(\frac{N_1}{N_2} \right)^2 \frac{1}{\mu_r} = \left(\frac{600}{200} \right)^2 \times \frac{1}{2} = 4.5$$

3. Ans (4)

After moving the loop about side PQ through 120° , the normal on the side PS from the wire will be passing through midpoint C and the net magnetic flux through the loop in the new position will be zero.

$$q = \frac{\Delta\phi}{R}$$

Flux through area element dA in initial state.



$$d\phi = \vec{B} \cdot d\vec{A} = \frac{\mu_0 I}{2\pi x} a \cdot dx$$

$$\phi = \int_a^{2a} \frac{\mu_0 I a dx}{2\pi x} = \frac{\mu_0 I a}{2\pi} \ln(2)$$

$\therefore$ Charge flown through the square loop

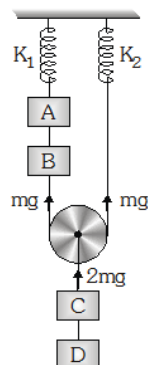
$$= \frac{\Delta\phi}{R} = \frac{\mu_0}{2\pi} \frac{I(a) \ln 2}{\ln 2} = \frac{\mu_0}{2\pi} I a = 4 \times 10^{-7} C$$

4. Ans (2)

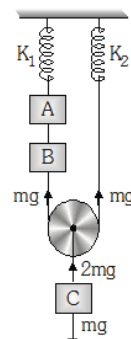
Initially :

$$K_2 x_{02} = mg$$

$$K_1 x_{01} = 3mg$$

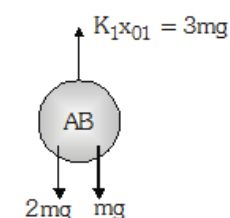


Just after the string is cut.



$$a_C = g \uparrow$$

This tension will not change spontaneously due to spring K_2



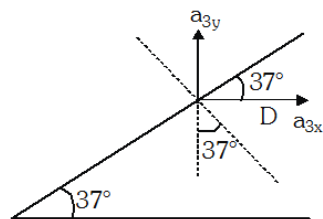
$$a_{A+B} = 0$$

Tension between A and B will not change.

$$T_{AB} = 2mg$$

5. Ans (5)

For C and D interface

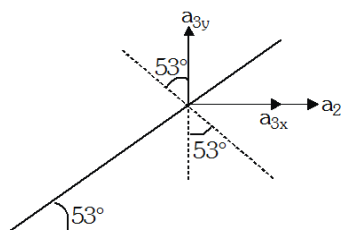


$$a_{3y} \cos 37^\circ = a_{3x} \times \sin 37^\circ$$

$$a_{3y} \left(\frac{4}{5} \right) = a_{3x} \left(\frac{3}{5} \right)$$

$$a_{3y} = \frac{3}{4} a_{3x}$$

for C and B interface



$$-a_{3y} \cos 53^\circ + a_{3x} \sin 53^\circ = a_2 \sin 53^\circ$$

$$-\left(\frac{3}{4} a_{3x} \right) \left(\frac{3}{5} \right) + a_{3x} \left(\frac{4}{5} \right) = 4 \left(\frac{4}{5} \right)$$

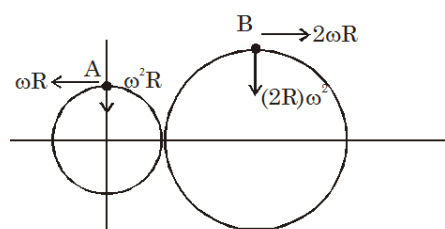
$$a_{3x} \left[\frac{-9}{20} + \frac{4}{5} \right] = \left[\frac{16}{5} \right]$$

$$a_{3x} \left(\frac{-9 + 16}{20} \right) = \left(\frac{16}{5} \right)$$

$$a_{3x} = \left(\frac{64}{7} \right)$$

$$a_{3y} = \frac{3}{4} \left(\frac{64}{7} \right) = \left(\frac{48}{7} \right)$$

6. Ans (9)



$$\text{At } t = \frac{\pi}{2\omega}$$

$$V_{B/A} = 3\omega R$$

$$a_{B/A} = R\omega^2$$

$$\text{R.O.C.} = \frac{(V_{B/A})^2}{(a_{B/A})_\perp} = \frac{9\omega^2 R^2}{R\omega^2} = 9R = 9$$

PART-2 : CHEMISTRY

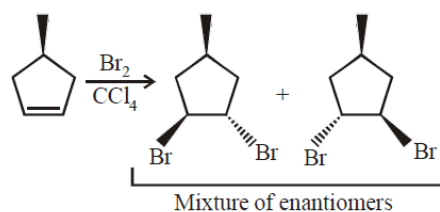
SECTION-I

1. Ans (A,B,D)

SbH₃ has higher bp than NH₃ due to vander waal forces.

In hydricids, the acidity increases down the group while for oxyacids, the acidity decreases down the group.

2. Ans (B,C)



4. Ans (A,D)

∴ For p orbital probability density at nucleus is zero.

∴ in depends of n. So option (A) is false.

∴ As per Bohr's model ground state is n = 1

∴ Angular momentum = $\frac{h}{2\pi}$, so option (B) is true.

For 1s orbital for radial probability distribution graph distance of antinode is inversely dependent on Z.

∴ $1 : \frac{1}{2} : \frac{1}{3} \Rightarrow 6 : 3 : 2$, so option (C) is true.

∴ As 'l' increases angular momentum increases but penetration power decreases. So option (D) is false.

5. Ans (A,B,D)

Rate of reaction increase substantially with increase in temperature since fraction of molecules crossing energy barrier increases.

$\Rightarrow k = Ae^{-E_a/RT}$ for all type of reaction (ΔH +ve or -ve) hence on $T \uparrow$, $k \uparrow$ (Rate $\uparrow$) for all type of reaction.

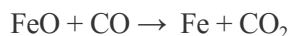
$\Rightarrow k = Ae^{-E_a/RT}$ if activation energy is zero than $k = A$ ∴ Rate of reaction remains unchanged if temperature changes.

6. Ans (B,C,D)

(A) At 500 – 800 K (lower temperature range in the blast furnace)



At 900 – 1500 K (higher temperature range in the blast furnace) :



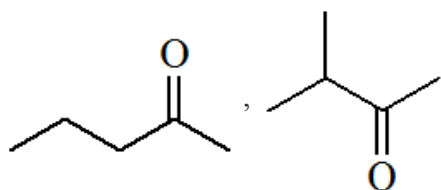
(B) calamine is ZnCO_3

(C) It contains Fe_2O_3 , SiO_2 and TiO_2 as impurities

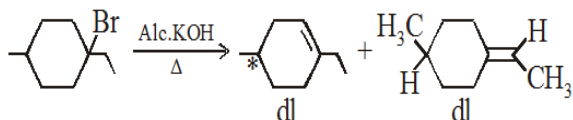
(D) The surface of solidified copper has blistered like appearances due to the evolution of SO_2 and so it is called blister copper.

SECTION-II

1. Ans (0.20)



2. Ans (4.00)



3. Ans (169.00)

a = 5 CO_2 , HNO_2 , HNO_3 , Benzene, H_2CO_3

b = 4 H_2SO_4 , HClO_4 , XeO_4 , H_3PO_4

4. Ans (1.26 to 1.27)

$\text{pH}_3 = \text{p}K_{a1} = 4.6$ (Best buffer of H_2A and HA^-)

$\text{pH}_4 = \text{p}K_{a2} = 8$ (Best buffer of HA^- and A^{2-})

$\text{pH}_1 = \frac{\text{p}K_{a1} + \text{p}K_{a2}}{2} = 6.3$ (Amphiprotic ion

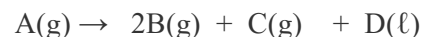
HA^- only)

Ratio = $\frac{\text{pH}_4}{\text{pH}_1} = \frac{8}{6.3} = 1.27$

5. Ans (18.00)

X = 16 , Y = 44 and Z = 2.

6. Ans (1171.50)



$$t = 0 \quad P_0$$

$$t = 30 \quad P_{0-x} \quad 2x \quad x \quad 31.5$$

$$t = 60 \quad P_{0-y} \quad 2y \quad y \quad 31.5$$

$$t = \infty \quad 0 \quad 2P_0 \quad P_0 \quad 31.5$$

$$P_0 - x + 2x + x + 31.5 = 931.5$$

$$\& 3P_0 + 31.5 = 1531.5 \Rightarrow P_0 = 500 \& x = 200$$

$$K = \frac{1}{30} \ln \left(\frac{P_0}{P_0 - x} \right) = \frac{1}{60} \ln \left(\frac{P_0}{P_0 - y} \right)$$

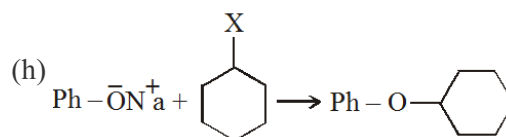
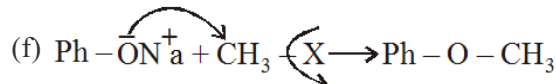
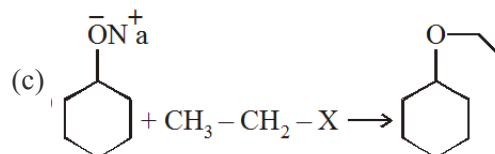
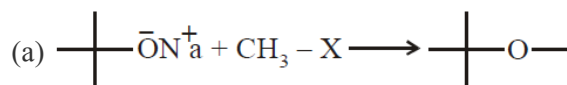
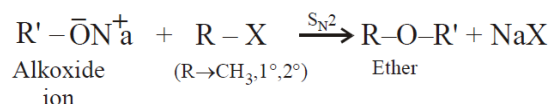
$$\Rightarrow \left(\frac{P_0}{P_0 - x} \right)^2 = \frac{P_0}{P_0 - y} = \left(\frac{500}{300} \right)^2$$

$$\frac{500}{500 - y} = \frac{25}{9} \Rightarrow y = 320$$

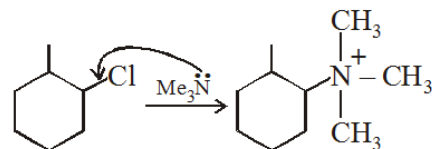
$$\text{Ans.} = P_{\text{Total}} = P_0 + 2y + 31.5 = 1171.5$$

SECTION-III

1. Ans (4)

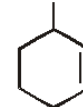


2. Ans (3)



Moist
 Ag_2O

Δ



Major product
(3 α H)

4. Ans (4)

$$K_{a1}K_{a2} = \frac{[H^+]^2 [S^{2-}]}{[H_2S]}$$

$$10^{-21} = \frac{(0.5)^2}{0.1} [S^{2-}]$$

$$\frac{1}{25} \frac{1}{25} \times 10^{-20} = [S^{2-}]$$

$$[S^{2-}] = 4 \times 10^{-22} \text{ M}$$

5. Ans (2)

	Metal	Ore	Chemical Formula	Reduction process
(i)	Al	Bauxite	$Al_2O_3 \cdot 2H_2O$	Hall Heroult process
(ii)	Cu	Copper pyrite	$CuFeS_2$	Self reduction in Bessemer converter
(iii)	Fe	Haemetite	Fe_2O_3	Blast Furnace
(iv)	Pb	Galena	PbS	self reduction

6. Ans (2)

$$\frac{A_1}{A_2} = \frac{\lambda_1 N_1}{\lambda_2 N_2}$$

$$\frac{\left(N_0 e^{-\frac{\ell n 2}{2} \times 2}\right) \frac{\ell n 2}{2}}{\left(N_0 e^{-\frac{\ell n 2}{4} \times 2}\right) \frac{\ell n 2}{4}} = \frac{\frac{1}{2} \times 2}{\frac{1}{\sqrt{2}}} = \sqrt{2}$$

PART-3 : MATHEMATICS

SECTION-I

1. Ans (A,B)

$$\frac{y^2 dx - 2xy dy}{x^2} = y^3 (x dy + y dx)$$

$$\Rightarrow -d\left(\frac{y^2}{x}\right) = y^3 d(xy)$$

$$\Rightarrow -\frac{d(y^2/x)}{y^2/x} = xy d(xy)$$

$$\therefore -\ln\left(\frac{y^2}{x}\right) = \frac{(xy)^2}{2} + c$$

$$\Rightarrow \ln\left(\frac{x}{y^2}\right) = \frac{(xy)^2}{2} + c$$

$$\Rightarrow \frac{x}{cy^2} = e^{(xy)^2/2}$$

2. Ans (B,D)

$$I_n = \int_0^{\pi/2} (\sin x + \cos x)^{n-1} d(\sin x - \cos x)$$

$$\Rightarrow I_n = \left[(\sin x + \cos x)^{n-1} \cdot (\sin x - \cos x) \right]_0^{\pi/2} + \int_0^{\pi/2} (n-1) (\sin x + \cos x)^{n-2} (\sin x - \cos x)^2 dx$$

$$\Rightarrow I_n = 2 + (n-1) \int_0^{\pi/2} (\sin x + \cos x)^{n-2} (1 - \sin 2x) dx$$

$$\Rightarrow I_n = 2 + (n-1) \int_0^{\pi/2} (\sin x + \cos x)^{n-2} (2 - (\sin x + \cos x)^2) dx$$

$$\Rightarrow I_n = 2 + 2(n-1) I_{n-2} - (n-1) I_n$$

$$\Rightarrow nI_n - 2(n-1) I_{n-2} = 2 \quad \forall n \geq 3$$

$$\text{Put } n = 3, \quad 3I_3 - 4I_1 = 2 \text{ where } I_1 = 2$$

$$\Rightarrow I_3 = \frac{10}{3}$$

3. Ans (A,B,C)

P - 5 & P - 6 (King & Queen)

4. Ans (A,B,C)

Let equation of tangents

$$\Rightarrow y = mx \pm \sqrt{2m^2 + 1}$$

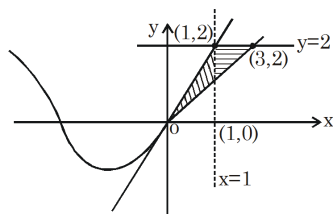
$$\sqrt{5} = \left| \frac{2\sqrt{2m^2 + 1}}{\sqrt{1 + m^2}} \right|$$

$$\Rightarrow m = \pm \frac{1}{\sqrt{3}}$$

$$y = \frac{\pm x \pm \sqrt{5}}{\sqrt{3}}$$

$$\text{Area} = \left| \frac{\frac{2\sqrt{5}}{\sqrt{3}} \cdot \frac{2\sqrt{5}}{\sqrt{3}}}{\frac{2}{\sqrt{3}}} \right| = \frac{10}{\sqrt{3}} \text{ sq. unit}$$

5. Ans (B,C,D)



$$A_1 = \int_0^1 \left(\frac{(2x+1)^2 - 1}{4} - x \right) dx$$

$$A_1 = \frac{1}{3}$$

$$A_2 = \frac{1}{2}$$

6. Ans (A,B,D)

$$I_1 = \int_1^e x \left(\frac{1+x}{x} \right) (x + \ln x)^{100} dx$$

$$= \frac{1}{101} \int_1^e x \cdot d(x + \ln x)^{100}$$

$$\therefore I_1 = \frac{1}{101} [x \cdot (x + \ln x)^{101}]^e - \frac{1}{101} \int_1^e (x + \ln x)^{101} dx$$

$$\Rightarrow I_1 = \frac{e \cdot (e+1)^{101} - 1}{101} - \frac{1}{101} \int_1^e (x + \ln x)^{101} dx$$

$$I_2 = \frac{1}{e} \int_1^e (t + \ln t)^{101} dt, \text{ put } e^{\sin x} = t$$

$$\text{thus } I_1 = \frac{e(e+1)^{101} - 1}{101} - \frac{e}{101} I_2$$

$$\Rightarrow I_1 = \frac{e}{101} I_2 = \frac{e(e+1)^{101} - 1}{101}$$

$$\therefore k = 1$$

SECTION-II

1. Ans (2.00)

$$\int_0^2 x f(x) dx = f(x) \cdot \frac{x^2}{2} \Big|_0^2 - \int_0^2 f'(x) \cdot \frac{x^2}{2} dx = 2$$

2. Ans (1.60)

$$\text{Curves are } y = x + x\sqrt{x}, x - x\sqrt{x}$$

$$\text{Required Area} = \int_0^1 (x + x\sqrt{x}) - (x - x\sqrt{x}) dx$$

$$= 2 \int_0^1 x^{3/2} dx$$

$$= \frac{4}{5} \text{ unit}^2$$

3. Ans (1.00)

$$-\frac{1}{(dy/dx)} = \frac{1}{2x} \Rightarrow dy + 2x dx = 0$$

$$\therefore y + x^2 = C$$

Since curve passes through (4, 3) $\therefore c = 19$.

4. Ans (1.00)

$$V = \frac{4}{3} \pi r^3 \Rightarrow \frac{dv}{dr} = 4\pi r^2$$

$$A = 4\pi r^2 \Rightarrow \frac{dA}{dr} = 4\pi r$$

$$\frac{dv}{dA} = \frac{r}{2} \Rightarrow \left(\frac{dv}{dA} \right)_{r=2} = 1$$

5. Ans (0.50)

Third term of GP : $ar^2 = 4$

$$A_n = 2 \left[\int_0^{a_n/2} 2\sqrt{a_n} \sqrt{x} dx + \int_{a_n/2}^{a_n} 2\sqrt{a_n} \sqrt{a_n - x} dx \right]$$

$$\therefore A_n = \frac{4\sqrt{2}}{3} a_n^2$$

$$A_3 = \frac{4\sqrt{2}}{3} \times 16 = \frac{64\sqrt{2}}{3}$$

6. Ans (2.50)

The perpendicular from focus to the tangent meet at the tangent at the vertex.

$$\therefore \text{Equation of perpendicular } 4x - 3y = 0$$

Since (3, 4) satisfies $\Rightarrow (3, 4)$: vertex and $3x + 4y = 25$; Tangent at the vertex.

$\therefore$ Required AB : length of focal chord of parabola distance from focus to tangent at the vertex = 5.

$$\therefore \text{Length of latus rectum} = 20 = AB.$$

SECTION-III

1. **Ans (8)**

$$I_2 = \frac{1}{2} \int_0^{1/2} \ln \sqrt{t} \cdot \ln(1-t) dt \text{ put } x^2 = t$$

$$= \frac{1}{4} \int_0^{1/2} \ln t \cdot \ln(1-t) dt$$

$$I_1 = 2 \int_0^{1/2} \ln x \cdot \ln(1-x) dx$$

$$\therefore \frac{I_1}{I_2} = 8$$

2. **Ans (2)**

$$x dy - y dx = \frac{x^4}{y} dx$$

$$\frac{y}{x} \left(\frac{x dy - y dx}{x^2} \right) = x dx$$

$$\Rightarrow \int \frac{y}{x} d \left(\frac{y}{x} \right) = \int x dx$$

$$\left(\frac{y}{x} \right)^2 = x^2 + 2c$$

$$\Rightarrow y = \sqrt{x^4 + 2cx^2}$$

$$\text{put } x = 1 \text{ to get } c = \frac{3}{2}$$

$$\therefore f(x) = \sqrt{x^4 + 3x^2}$$

$$f(-1) = 2$$

3. **Ans (2)**

$$\frac{dy}{dx} = \frac{-2\cos^{-1}x}{\sqrt{1-x^2}}$$

$$\Rightarrow \sqrt{1-x^2} \cdot y_1 = -2\cos^{-1}x$$

$$\therefore \sqrt{1-x^2} \cdot y_2 - \frac{2x \cdot y_1}{2\sqrt{1-x^2}} = \frac{2}{\sqrt{1-x^2}}$$

$$\Rightarrow (1-x^2)y_2 - x \cdot y_1 = 2$$

4. **Ans (3)**

SP, 2a, SQ are in HP where $4a = 8$

5. **Ans (7)**

$$\text{Ellipse : } \frac{x^2}{3} + \frac{y^2}{2} = 1$$

$$\text{Condition of tangency : } 4h^2 = 3k^2 + 2$$

$$\text{Required locus : } 4x^2 = 3y^2 + 2$$

$$\Rightarrow 4x^2 - 3y^2 = 2$$

$$\text{Thus : } b^2 = a^2(e^2 - 1), \text{ where } a^2 = \frac{1}{2}, b^2 = \frac{2}{3}$$

$$\frac{2}{3} = \frac{1}{2}(e^2 - 1) \Rightarrow e^2 = \frac{7}{3}$$

6. **Ans (4)**

$$\int e^x \left(\tan^{-1}x + \frac{2x}{(1+x^2)^2} \right) dx$$

$$= \int e^x \left(\tan^{-1}x + \frac{1}{1+x^2} - \frac{1}{1+x^2} + \frac{2x}{(1+x^2)^2} \right) dx$$

$$= e^x \left(\tan^{-1}x - \frac{1}{1+x^2} \right) + c$$

$$\therefore g(x) = -\frac{1}{1+x^2}$$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Main)
ALLEN COMMON TEST
31-10-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	A	A	C	A	A	B	B	A	A
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	C	B	A	B	B	D	C	A	C	D
	Q.	1	2	3	4	5	6	7	8	9	10
A.	6.00	4.00	1.73	4.00	3.00	6.00	3.33	0.25	2.50	1.00	

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	B	C	C	B	D	A	B	A	D
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	C	D	C	A	C	C	A	C	D	A
	Q.	1	2	3	4	5	6	7	8	9	10
A.	4.70	6.00	17.00	200.00	3.00	5.00	3.01	6.00	300.00	1.00	

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	B	B	A	C	B	B	C	A	D
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	C	D	B	C	A	D	C	C	D	D
	Q.	1	2	3	4	5	6	7	8	9	10
A.	2.00	4.00	1.00	3.00	2.00	16.00	7.00	6.00	1.00	500.00	

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

1. Ans (B)

From figure it is clear

$$\alpha = 180 - (i + r)$$

$$\text{and } \mu = \frac{\sin i}{\sin r} \Rightarrow \sqrt{3} = \frac{\sin 60^\circ}{\sin r}$$

$$r = 30^\circ$$

$$\alpha = 90^\circ$$

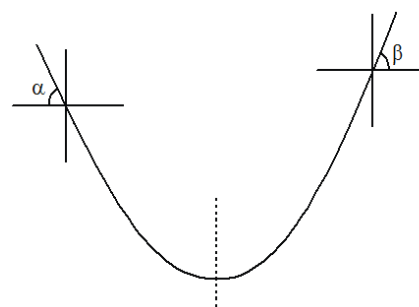
2. Ans (A)

$$i = \sqrt{3}A$$

$$\frac{q_m^2}{2C} = \frac{q^2}{2C} + \frac{1}{2}Li^2 \Rightarrow q_{\max} = 6C$$

3. Ans (A)

$$T_1 \cos \alpha = T_3 = T_2 \cos \beta$$

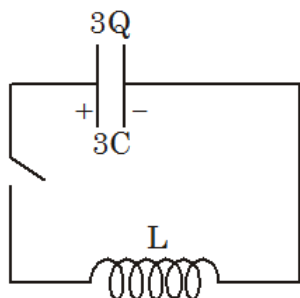


$$T_1 \sin \alpha + T_2 \sin \beta = mg$$

$$\Rightarrow mg = T_1 \sqrt{1 - \left(\frac{T_3}{T_1}\right)^2} + T_2 \sqrt{1 - \left(\frac{T_3}{T_2}\right)^2}$$

$$\Rightarrow m = \frac{\sqrt{T_1^2 - T_3^2} + \sqrt{T_2^2 - T_3^2}}{g}$$

4. **Ans (C)**



The equivalent circuit is as shown in fig.

$$U_{L(\max)} = U_{C(\max)} = \frac{1}{2} \frac{(3Q)^2}{3C} = \frac{3Q^2}{2C}$$

5. **Ans (A)**

Voltage across capacitor will lag current by 90°

$$\therefore i_C = \frac{200\sqrt{2}}{10} \sin\left(\omega t + \frac{\pi}{4} + \frac{\pi}{2}\right)$$

$$i = 20\sqrt{2} \sin\left(\omega t + \frac{3\pi}{4}\right)$$

$\therefore$ Correct answer is (A)

6. **Ans (A)**

Initial acceleration of block is zero so tension in thread should be equal to Mg .

7. **Ans (B)**

We use $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$, with two different focal lengths to calculate the two different v 's. The two extreme focal length can be determined from the two extreme values of μ .

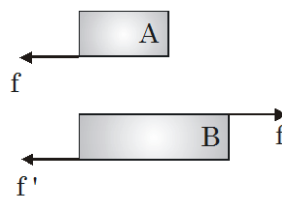
8. **Ans (B)**

$$\text{Shift } dS = \left(1 - \frac{1}{\mu}\right) dx$$

$$\int ds = \int_0^t \left(1 - \frac{t}{\mu_0(t+x)}\right) dx$$

$$\Rightarrow t - t[\ln(t+x)]_0^t = t - t\ln 2 = t(1 - \ln 2)$$

9. **Ans (A)**



Since $\frac{\mu}{4} < \frac{\mu}{2} < \mu$, the upper block will move faster than the middle block and hence force of friction on upper block is towards left

$$\therefore f = \frac{\mu}{4} mg$$

$$\therefore a_A = \frac{\mu g}{4} (\text{leftward})$$

$$f' = \frac{\mu}{2} \left(m + \frac{m}{2}\right) g = \frac{3}{4} \mu mg$$

$$\therefore a_B = \mu g (\text{leftward})$$

$$\therefore a_{A/B} = \frac{3}{4} \mu g (\text{rightward})$$

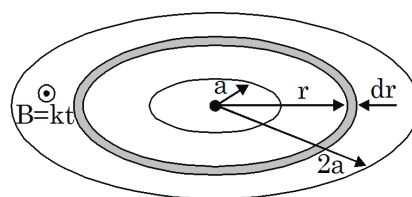
For sliding down the block B, A has to move a distance $\frac{3L}{8}$ relative to block B.

$$\frac{3L}{8} = \frac{1}{2} \frac{3}{4} \mu g t^2$$

$$\therefore t = \sqrt{\frac{L}{\mu g}}$$

10. **Ans (A)**

The emf induced in a thin circular ring of radius 'r' is



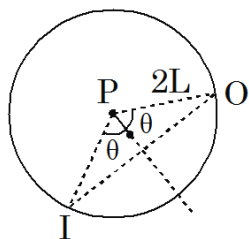
$$\varepsilon = \left| \frac{d\Phi}{dt} \right|$$

$$\varepsilon = \frac{d}{dt} (kt\pi r^2) = k\pi r^2$$

$$\int dl = \int_a^{2a} \frac{k\pi r^2 h dr}{\rho 2\pi r}$$

$$l = \frac{kh}{2\rho} \left[\frac{r^2}{2} \right]_a^{2a} = \frac{3kha^2}{4\rho}$$

11. Ans (C)



Radius = $2L$

$\theta \rightarrow \theta - \Delta\theta$

$\angle OPI = 2\theta - 2\Delta\theta$

So, angular velocity of image = 2ω (ACW)

I always moves on the circle.

12. Ans (B)

$u = -x$

$y = -y$ (real image)

$f = -z$

$$\frac{1}{x} + \frac{1}{y} = \frac{1}{z}$$

13. Ans (A)

The eye is least strained if the final image is formed at infinity. In such a case, the image formed by the objective should fall at the focus of that eyepiece. As $f_e = 5$ cm and $L = 7$ cm, this first image should be formed at 7 cm, this first image should be formed at 7 cm – 5 cm, this first image should be formed at 7 cm – 5 cm = 2 cm from the objective. Thus, $u = +2$ cm. Also, $f_o = 0.5$ cm. For the objective, using

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f_o}; \quad \frac{1}{u} = \frac{1}{v} - \frac{1}{f_o} = \frac{1}{2\text{cm}} - \frac{1}{0.5\text{cm}}$$

$$\text{or } u = -\frac{2}{3}\text{cm}$$

The angular magnification in this case is

$$m = \frac{vD}{uf_e} = \frac{2\text{cm}}{-(2/3)\text{cm}} \frac{25\text{cm}}{5\text{cm}} = -15$$

14. Ans (B)

Rms current through resistor ' R_1 ' is

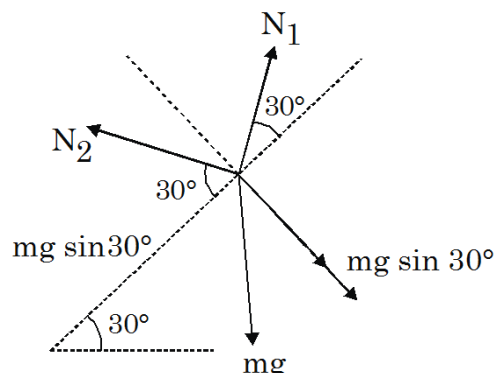
$$I_1 (\text{rms}) = \frac{50}{\sqrt{2} \times 10\sqrt{2}} = \frac{5}{2} \text{ A}$$

$\therefore$ Average power developed in the resistor ' R_1 ' is

$$P_1 = I_{1(\text{rms})}^2 R_1 = \left(\frac{5}{2}\right)^2 10 = \frac{250}{4} = 62.50 \text{ W}$$

15. Ans (B)

$$\frac{\sqrt{3}N_1}{2} - \frac{mg}{2} - \frac{N_2\sqrt{3}}{2} = ma$$



$$\frac{\sqrt{3}N_1}{2} - \frac{\sqrt{3}N_2}{2} = ma + \frac{mg}{2} \dots (i)$$

$$\Rightarrow \frac{N_1 + N_2}{2} = \frac{mg\sqrt{3}}{2}$$

$$\Rightarrow \frac{\sqrt{3}N_1}{2} + \frac{\sqrt{3}N_2}{2} = \frac{3mg}{2} \dots (ii)$$

From equation (i) and (ii), we have

$$\sqrt{3}N_1 = 2mg + ma, \quad \sqrt{3}N_2 = mg - ma$$

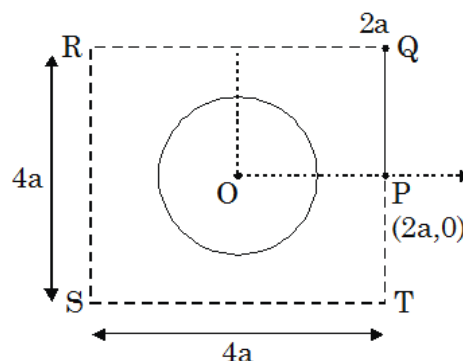
Since cylinder does not lose the contact, so

$$N_2 \geq 0$$

$$\Rightarrow a \leq g \Rightarrow a_{\text{max}} = g$$

16. Ans (D)

$$\int_P^Q \vec{E} \cdot d\vec{y} = \frac{1}{8} \left[\vec{E} \cdot d\vec{l} \text{ for square TQRST} \right]$$



$$= \frac{1}{8} \frac{d}{dt} [\pi a^2 B_0 t]$$

$$= \frac{\pi}{8} B_0 a^2$$

17. Ans (C)

$$\frac{1}{v} + \frac{1}{d} = \frac{1}{f} \Rightarrow v = \frac{fd}{d-f}$$

For second and reflection object

$$u = d - v = \frac{d^2 - 2df}{d-f}$$

location on final image

$$\frac{1}{v_1} + \frac{1}{u} = \frac{1}{f}$$

$$v_1 = \frac{fu}{u-f} = \frac{f \left(\frac{d^2 - 2df}{d-f} \right)}{\frac{d^2 - 2df}{d-f} - f}$$

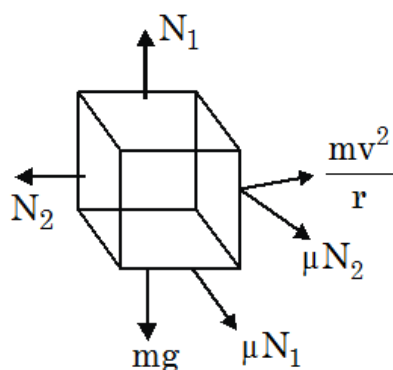
location of final image at hole $v_1 = d$

$$d_2 - 4df - 3f^2 = 0$$

So $d = f$ and $3f$

18. Ans (A)

F.B.D of block at any instant



$$\mu(N_1 + N_2) = -mv \frac{dv}{dx} \Rightarrow \mu(\cancel{mg}) + \frac{mv^2 \mu}{r}$$

$$\Rightarrow \mu \left(\frac{rg + v^2}{r} \right) = -v \frac{dv}{dx}$$

$$\int_0^{2\pi r} \frac{\mu}{r} dx = \int_{v_0}^0 -\frac{v dv}{(rg + v^2)} \left(rg + v^2 = t, 2v dv = dt, v dv = \frac{dt}{2} \right)$$

$$\int \frac{\mu}{r} dx = -\int \frac{dt}{t.2} \Rightarrow \frac{\mu}{r} \times 2\pi r = -\frac{1}{2} \left[\ln \frac{rg}{rg + v_0^2} \right]$$

$$\mu.4\pi = \ln \left(\frac{rg + v_0^2}{rg} \right)$$

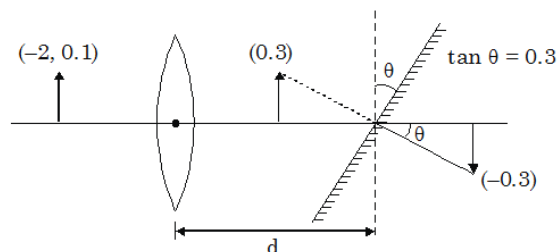
$$\Rightarrow rg + v_0^2 = rge^{4\pi\mu} \Rightarrow v_0 = \sqrt{rg(e^{4\pi\mu} - 1)}$$

19. Ans (C)

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{v} + \frac{1}{2} = \frac{10}{15}$$

$$\Rightarrow \frac{1}{v} = \frac{2}{3} - \frac{1}{2} = \frac{4-3}{6} = \frac{1}{6}$$

$$\Rightarrow v = 6 \text{ meter}$$



$$m = \frac{v}{u} = -3$$

$$\tan \theta = \frac{0.3}{x} \Rightarrow 0.3 = \frac{0.3}{x} \Rightarrow x = 1$$

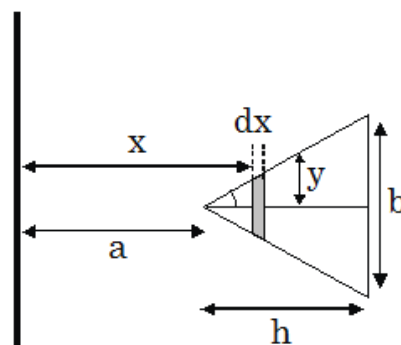
$$x = 6 - d \Rightarrow 6 - d = 1$$

$$\therefore d = 5m$$

20. Ans (D)

$$d\phi = \left(\frac{\mu_0 i}{2\pi x} \right) \cdot \left(\frac{b}{h} (x-a) dx \right)$$

$$\int d\phi = \int_a^{a+h} \frac{\mu_0 i}{2\pi} \left[\frac{b}{h} \left[1 - \frac{a}{x} \right] \right] dx$$



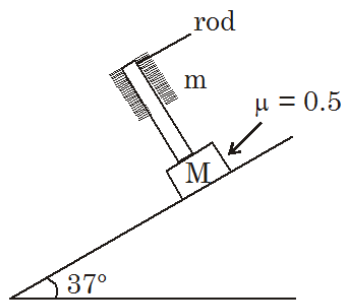
$$\phi = \frac{\mu_0 i b}{2\pi h} [x - a \ln x]_a^{a+h}$$

$$\phi = \frac{\mu_0 i}{2\pi h} [h - a \ln 2] = \frac{4\pi \times 10^{-7}}{2\pi} \times 2 \times [1 - \ln 2]$$

$$\therefore 4 \times 10^{-2} [1 - \ln 2]$$

SECTION-II

1. Ans (6.00)



For equilibrium of block

$$Mg \sin \theta = \mu (N)$$

$$= \mu (Mg \cos \theta + mg \cos \theta)$$

Solving it we get $m = 6 \text{ kg}$

2. Ans (4.00)

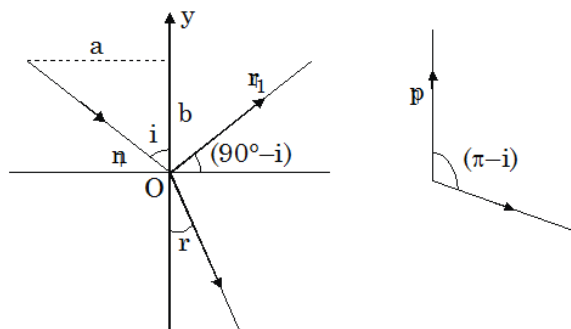
$$i_{\text{rms}} = \sqrt{\frac{\int_0^t i^2 dt}{\int_0^t dt}}$$

$$= \sqrt{\frac{\int_0^1 16 dt + \int_1^2 (4 - 4t)^2 dt}{\int_0^2 (dt)}}$$

$$= 4\sqrt{\frac{2}{3}} \text{ Amp.}$$

3. Ans (1.73)

$$\hat{p} = \hat{j}, \hat{n} = a\hat{i} - b\hat{j} = \frac{1}{2}\hat{i} - \frac{\sqrt{3}}{2}\hat{j}$$



$$\hat{r}_1 = \hat{n} - 2(\hat{n} \cdot \hat{p})\hat{p}$$

$$= (a\hat{i} - b\hat{j}) - 2(-b)\hat{j} = a\hat{i} + b\hat{j}$$

$$\hat{n} \cdot \hat{p} = \cos(\pi - i) = -b$$

$$\cos i = b = \frac{\sqrt{3}}{2} \Rightarrow i = 30^\circ$$

$$\mu = \cot i = \sqrt{3}$$

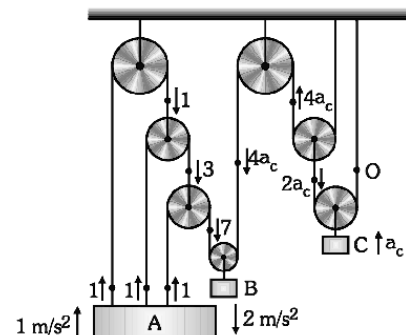
4. Ans (4.00)

The Half of the maximum current is equal to

$$i_1 = \frac{B\omega l^2}{4R} \text{ at } t = \frac{L}{R} \ln 2$$

$$\text{The torque at this instant} = \frac{B^2 \omega l^4}{8R} = 4N - m$$

5. Ans (3.00)



$$2 \times 2 = 7 + 4 a_c$$

$$a_c = -\frac{3}{4}$$

$$\vec{a}_c = \frac{3}{4} \text{ m/s}^2 \downarrow$$

6. Ans (6.00)

$$i_L = 3A; I_C = 4.5 A$$

$$i = 1.5 A = i_C - i_L$$

$$= \frac{90}{1.5} = 60\Omega$$

7. Ans (3.33)

$$P = -\frac{2}{R} \dots (1)$$

$$P' = \frac{P}{3} = -\frac{2}{R} + (\mu - 1)\frac{2}{R} \dots (2)$$

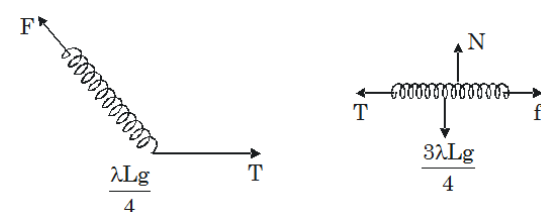
Using (1) & (2)

$$-\frac{2}{3R} = \frac{2}{R}(\mu - 2)$$

$$-\frac{1}{3} = \mu - 2 \Rightarrow \mu = \frac{5}{3}$$

8. Ans (0.25)

$$F \cos 37^\circ = \frac{\lambda L}{4} g$$



$$F \sin 37^\circ = T = f \leq \mu N \left(N = \frac{3\lambda Lg}{4} \right)$$

$$\Rightarrow \mu \geq 1/4, \Rightarrow \mu_{\text{min}} = 1/4$$

9. Ans (2.50)

$$X_C = \frac{V_C}{I} = \frac{1000}{1} = 1000\Omega$$

Current lags behind voltage

So, box has an inductor

$$\text{Power factor} = 0.8 = \frac{R}{Z} = \frac{800}{\sqrt{(800)^2 + (X_L - 1000)^2}}$$

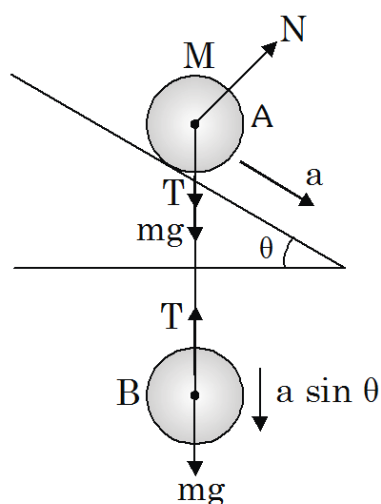
$$X_L = 1600 \Omega$$

$$\omega L = 1600$$

$$L = \frac{1600}{2 \times 3.2 \times 100} = 2.5H$$

10. Ans (1.00)

$$Mg \sin \theta + T \sin \theta = Ma$$



$$mg - T = ma \sin \theta$$

solving, we get

$$a = \frac{(M + m) g \sin^2 \theta}{M + m \sin^2 \theta}$$

PART-2 : CHEMISTRY

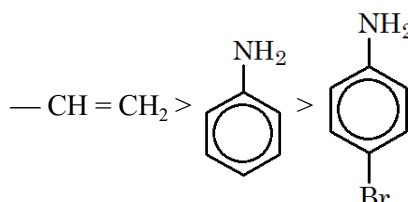
SECTION-I

1. Ans (B)

Minimum $pK_b \Rightarrow$ Maximum basic strength.

Basic strength: Localised lone pair $>$ Delocalised lone pair

Basic strength order : $(C_2H_5)_2NH > H_2N - CH_2$



2. Ans (B)

Dissociation of water is an endothermic reaction.

$K_w =$ Dissociation constant $= 10^{-14}$ at 25°C .

As $T \downarrow \Rightarrow K_w \downarrow \Rightarrow pK_w \uparrow$

$\Rightarrow$ pH of neutral water at $10^\circ\text{C} > 7$.

$\Rightarrow$ Solution at pH = 7 is slightly acidic.

3. Ans (C)

Blister copper is a impure form of copper, not an ore.

4. Ans (C)

$$K = Ae^{-E_a/RT}$$

$$\ln K = \ln A - \frac{E_a}{RT}$$

Comparing with $y = c + mx$

$$m = \frac{-E_a}{R} \text{ \& } c = \ln A$$

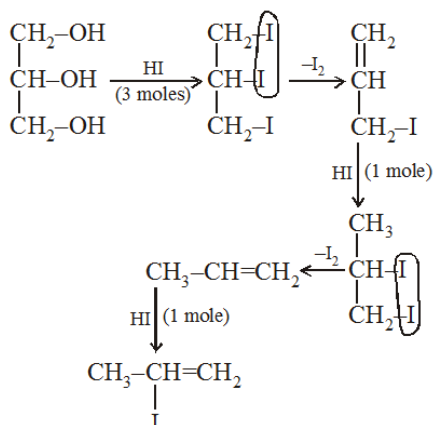
$\ln K(s^{-1})$	$\frac{1}{T}(\text{in } K^{-1})$	$T(\text{in } K)$
4	0.004	250K
5	0.003	333.33K
6	0.002	500K
7	0.001	1000K

$$\text{Slope} = \frac{-E_a}{R} = \frac{-1}{0.001}$$

$$\Rightarrow E_a = R \times 1000$$

$$= 2 \text{ Kcal/mole}$$

5. **Ans (B)**



6. **Ans (D)**

$$(A) \Rightarrow 1s + 1s = \sigma_{s-s}$$

$$(B) \Rightarrow 2p_x + 2p_x = \sigma_{p-p}$$

$$(C) \Rightarrow 2p_y + 2p_y = \pi_{p-p}$$

$$(D) \Rightarrow 2p_y - 2p_y = \pi^*_{p-p}$$

7. **Ans (A)**

Vapour density = 37 $\Rightarrow$ Molecular mass = 74

$\Rightarrow$ For one mole compound :

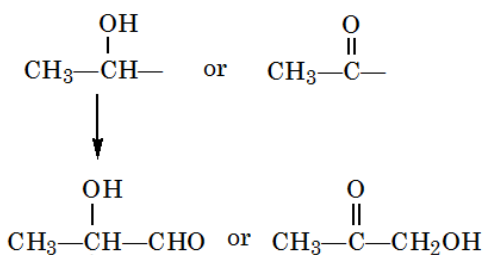
$$\text{Mass of Carbon} = \frac{48.68}{100} \times 74\text{g} \sim 36\text{g}$$

$$\text{Mass of Hydrogen} = \frac{8.11}{100} \times 74\text{g} \sim 6\text{g}$$

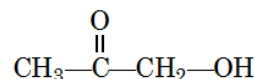
$$\text{Mass of Oxygen} = \frac{43.24}{100} \times 74\text{g} \sim 32\text{g}$$

$\Rightarrow$ Molecular formula : $\text{C}_3\text{H}_6\text{O}_2$

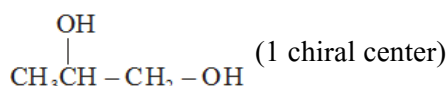
Iodoform test & Tollen's reagent test $\Rightarrow$



$\therefore$ No chiral atom present $\Rightarrow$



$\Rightarrow$ Product formed on reaction with NaBH_4 is



$\Rightarrow$ Compound undergoes aldol condensation.

$\Rightarrow$ Compound undergoes addition reaction with NaHSO_3 .

$\Rightarrow$ Acidic hydrogen is present attached to O.

8. **Ans (B)**

$$\frac{d[D]}{dt} = k[A] \Rightarrow \text{Rate of reaction} = k'[A]$$

$$\Rightarrow [D] = 3A_0 e^{-k't}$$

$$-\frac{d[A]}{dt} = \frac{1}{3} \frac{d[D]}{dt}$$

9. **Ans (A)**

$$(A) \text{ KE}_{\text{max}} = \frac{1240}{155} \text{ eV} - 2 \text{ eV} = 6 \text{ eV} ;$$

$$\lambda = \frac{h}{p} = \sqrt{\frac{150}{V}} \text{ \AA}$$

$$\text{for electron} = \sqrt{\frac{150}{6}} \text{ \AA} = 5 \text{ \AA} \quad (B)$$

$$\lambda = \frac{h}{p} = \frac{6.6 \times 10^{-34} \text{ J/s}}{6.6 \times 10^{-27} \text{ kg} \times 10^4 \text{ m/s}} = 10^{-11} \text{ m} = 0.1 \text{ \AA}$$

$$(C) \lambda = \frac{h}{p} = \frac{6.6 \times 10^{-34} \text{ J/s}}{66 \text{ kg} \times 6 \times \frac{5}{18} \text{ m/s}} = 6 \times 10^{-36} \text{ m}$$

$$(D) \lambda = \frac{h}{p} = \frac{6.6 \times 10^{-34} \text{ J/s}}{330 \times 10^{-3} \text{ kg} \times 150 \times \frac{5}{18} \text{ m/s}} = 4.8 \times 10^{-35} \text{ m}$$

(de-Broglie wavelength of macroscopic objects

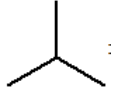
are small and non-significant as compared to

microscopic objects).

10. **Ans (D)**

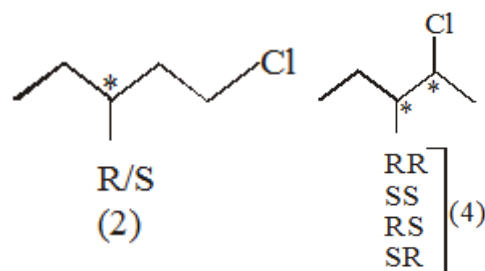
(A)  $\Rightarrow$ (6 sp^3 carbon atoms)

(B)  $\Rightarrow$ (5 sp^3 carbon atoms)

(C)  $\Rightarrow$ (4 sp^3 carbon atoms)

(D) $\text{H}_2\text{C}=\text{C}=\text{CH}_2 \Rightarrow$ (1 sp and 2 sp^2 carbon atoms)

11. **Ans (C)**



12. **Ans (D)**

$$\frac{1}{\lambda_1} = RZ^2 \left[\frac{1}{3^2} - \frac{1}{\infty^2} \right] \Rightarrow x = \frac{9}{4R} \text{Å}$$

$$\frac{1}{\lambda_2} = RZ^2 \left[\frac{1}{2^2} - \frac{1}{3^2} \right] \Rightarrow \lambda_2 = \frac{9}{5R} = \frac{4x}{5} \text{Å}$$

13. **Ans (C)**

(A) Basic strength: $\text{CH}_3\text{NH}_2 > \text{H}_2\text{O}$

(B) Basic strength: $\text{CH}_3\text{O}^- > \text{CH}_3\text{NH}_2$

(C) Acidic strength: $\text{HBr} > \text{CH}_3\text{O}^+\text{H}_2$

(D) Basic strength: $\text{OH}^- > \text{CH}_3\text{NH}_2$

The reaction favour the direction from SA & SB to WA & WB.

14. **Ans (A)**

$$250 \text{ ml of } 0.01 \text{ M HCl} \rightarrow \text{Moles of } \text{H}^+ = \frac{0.01}{4}$$

$$250 \text{ ml of } 0.02 \text{ M H}_2\text{SO}_4 \rightarrow \text{Moles of } \text{H}^+ = \frac{0.04}{4}$$

$$250 \text{ ml of } 0.03 \text{ M HClO}_3 \rightarrow \text{Moles of } \text{H}^+ = \frac{0.03}{4}$$

$$\text{Total moles of } \text{H}^+ \text{ in 1 litre solution} = 0.02$$

15. **Ans (C)**

1° Alcohol $\xrightarrow{\text{PCC}}$ Aldehyde

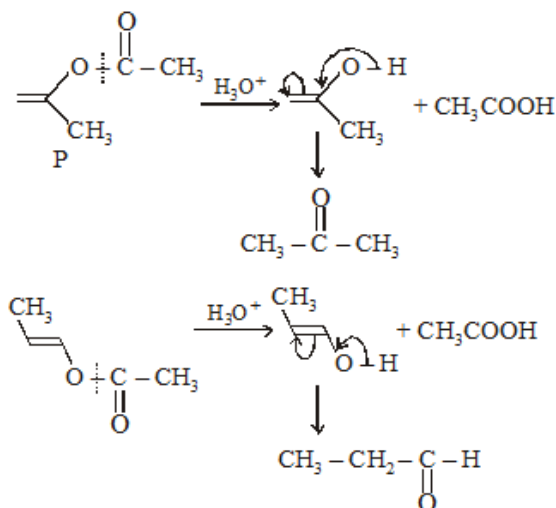
2° Alcohol $\xrightarrow{\text{PCC}}$ Ketone

3° Alcohol $\xrightarrow{\text{PCC}}$ ×

17. **Ans (A)**

Impurities from the blister copper deposit as anode mud which contains antimony, selenium, tellurium, silver, gold and platinum.

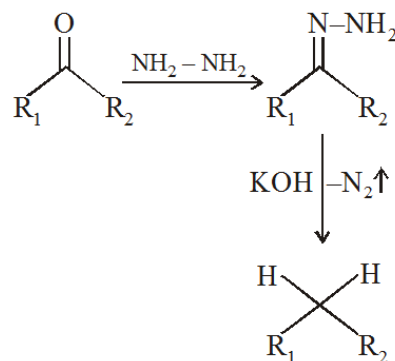
18. **Ans (C)**



19. **Ans (D)**

Theory based.

20. **Ans (A)**



SECTION-II

1. **Ans (4.70)**

$$\text{HA} \xrightleftharpoons{K_a} \text{H}^+ + \text{A}^- \quad K_a = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]}$$

$$\log K_a = \log[\text{H}^+] + \log \frac{[\text{A}^-]}{[\text{HA}]}$$

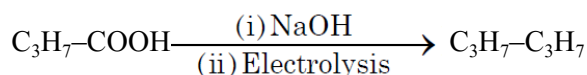
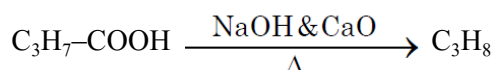
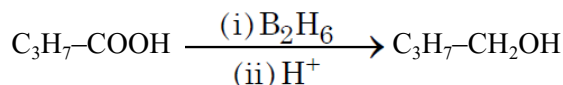
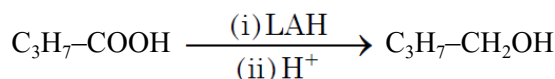
$$\text{If } [\text{A}^-] = [\text{HA}] \Rightarrow \text{p}K_a = \text{pH}$$

$$\Rightarrow \text{pH} = 5 - \log 2 \Rightarrow \text{pH} = 4.70$$

2. **Ans (6.00)**

Non polar molecules: BeH_2 , CO_2 , H_2 , BF_3 , CH_4 , CCl_4

3. **Ans (17.00)**



$$\text{Number of carbon atoms} = A + B + C + D \Rightarrow 4$$

$$+ 4 + 3 + 6 = 17$$

4. **Ans (200.00)**

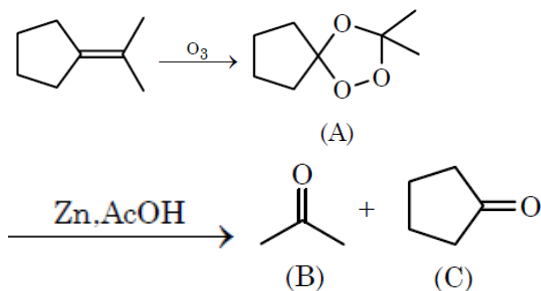
$$N = N_0 e^{-\lambda t}$$

$$\ln \frac{N}{N_0} = -\lambda t$$

$$t = \frac{1}{\lambda} \ln \frac{N_0}{N}$$

$$= \frac{30}{\ln 2} \ln 100 = 30 \frac{\log 100}{\log 2} = \frac{30 \times 2}{0.3} = 200 \text{ years}$$

6. **Ans (5.00)**



$$\text{Total oxygen atoms} = 3 + 1 + 1 = 5$$

7. **Ans (3.01)**

$$k = \frac{1}{t} \ln \frac{a_t}{a_0}$$

$$= \frac{1}{10} \ln 10 \text{ h}^{-1}$$

$$\text{Generation time} = \frac{1}{k} \ln 2$$

$$= 10 \frac{\ln 2}{\ln 10} = 10 \log 2 = 3.01 \text{ hours.}$$

8. **Ans (6.00)**

(i), (iv), (vii), (ix), (x) and (xi) are aromatic.

9. **Ans (300.00)**

MO can be reduced by C at 2000°C.



$$|\Delta G_r| = 300 \text{ kJ/mole}$$

10. **Ans (1.00)**

$$T.E. = -13.6 \frac{Z^2}{n^2} \text{ eV}$$

For 2nd orbit in He⁺ ion $\rightarrow T.E. = -13.6 \text{ eV}$

For 2nd E.S. of Li²⁺ ion $\rightarrow T.E. = -13.6 \text{ eV}$

$$K.E. = |T.E| = 13.6 \text{ eV}$$

PART-3 : MATHEMATICS

SECTION-I

1. **Ans (C)**

Let P(h, k) be the point from which two

tangent are drawn to $y^2 = 12x$

Any tangent to $y^2 = 12x$ is

$$T : y = mx + \frac{3}{m}$$

$$m^2 h - mk + 3 = 0$$

$$m_1 + m_2 = \frac{k}{h} ; m_1 m_2 = \frac{3}{h}$$

$$m_1 = 2m_2 \text{ (given)}$$

$$\therefore 3m_3 = \frac{k}{h} \Rightarrow m_3 = \frac{k}{3h}$$

$$2m_2^2 = \frac{3}{h} \Rightarrow 2 \cdot \frac{k^2}{9h^2} = \frac{3}{h} \Rightarrow y^2 = \frac{27}{2} x$$

2. **Ans (B)**

$$\int_{\pi/4}^{\pi/2} [|\sin x| + |\cos x| + [\operatorname{cosec} x]] dx$$

$$\int_{\pi/4}^{\pi/2} [|\sin x| + |\cos x|] dx + \int_{\pi/4}^{\pi/2} [\operatorname{cosec} x] dx$$

$$= \int_{\pi/4}^{\pi/2} 1 dx + \int_{\pi/4}^{\pi/2} 1 dx$$

$$1 < |\sin x| + |\cos x| \leq \sqrt{2}$$

$$= \pi/2$$

3. **Ans (B)**

$$I =$$

$$\int_0^{\pi/4} (\cos x - \sin x) dx + \int_{\pi/4}^{5\pi/4} (\sin x - \cos x) dx + \int_{5\pi/4}^{2\pi} (\cos x - \sin x) dx$$

$$= 4\sqrt{2}$$

4. Ans (A)

$$I = \int_0^a \ln(1 + \tan a \tan x) dx \quad \dots(1)$$

$$I = \int_0^a (\ln(1 + \tan a \tan(a-x))) dx$$

$$= \int_0^a \ln\left(1 + \frac{(\tan a(\tan a - \tan x))}{1 + \tan a \tan x}\right) dx =$$

$$\int_0^a \ln\left(\frac{1 + \tan^2 a}{1 + \tan a \tan x}\right) dx \quad \dots(2)$$

$$2I = \int_0^a \ln(\sec^2 a) dx = a \cdot \ln \sec^2 a$$

$$I = a \ln(\sec a)$$

$$\frac{d}{da} (a \ln \sec a) = \ln(\sec a) + a \left(\frac{\sec a}{\sec a}\right) \tan a$$

$$= a \tan a + \ln \sec a.$$

5. Ans (C)

The distance between the foci is 6, so $ae = 3$.

The sum of the distances from (4, 2) to each of the foci is equal to major axis length,

$$\text{so } 2a = \sqrt{(4-2)^2 + (2-4)^2} + \sqrt{(4-2)^2 + (2+2)^2} =$$

$$\sqrt{4+4} + \sqrt{4+16} = \sqrt{8} + \sqrt{20}$$

$$= 2\sqrt{2} + 2\sqrt{5} \Rightarrow a = \sqrt{2} + \sqrt{5}$$

$$\text{Also, for an ellipse, } b^2 = a^2 - a^2 e^2 = (\sqrt{2} + \sqrt{5})^2$$

$$- 3^2 = 7 + 2\sqrt{10} - 9 = -2 + 2\sqrt{10}.$$

$$\text{Thus, we have } (ab)^2 = (7 + 2\sqrt{10})(-2 + 2\sqrt{10})$$

$$= -14 + 14\sqrt{10} - 4\sqrt{10} + 40 = 26 + 10\sqrt{10}$$

6. Ans (B)

$$I_1 = \int_0^x e^{t(x-t)} dt$$

$$t(x-t) = -[t^2 - tx] = -\left[\left(t - \frac{x}{2}\right)^2 - \frac{x^2}{4}\right] =$$

$$\frac{x^2}{4} - \left(t - \frac{x}{2}\right)^2$$

$$I_1 = \int_0^x e^{\frac{x^2}{4} - \left(t - \frac{x}{2}\right)^2} dt = e^{\frac{x^2}{4}} \int_0^x e^{-\frac{(2t-x)^2}{4}} dt ; 2t$$

$$-x = y \Rightarrow dt = \frac{dy}{2}$$

$$= \frac{e^{x^2/4}}{2} \int_{-x}^x e^{-\frac{y^2}{4}} dy$$

$$I_1 = e^{\frac{x^2}{4}} \int_0^x e^{-\frac{t^2}{4}} dt = e^{\frac{x^2}{4}} \cdot I_2 \Rightarrow \frac{I_1}{I_2} = e^{\frac{x^2}{4}}$$

7. Ans (B)

$$\text{Let } I = \int_{-3}^3 \frac{\ln|x| dx}{1 + e^x}$$

$$= \int_{-3}^3 \frac{\ln|x| dx}{1 + e^{-x}} \left(\int_{-a}^a f(x) dx = \int_{-a}^a f(-x) dx \right)$$

$$\Rightarrow 2I = \int_{-3}^3 \ln|x| dx = 2 \int_0^3 \ln|x| dx$$

$$\Rightarrow I = (x(\ln x - 1))_0^3 = 3(\ln 3 - 1)$$

8. **Ans (C)**

$$\begin{aligned} & \int (2 - x \tan x) \sqrt{\cos x} dx \\ &= \int (2\sqrt{\cos x} - x \tan x \sqrt{\cos x}) dx \\ &= \int (f(x) + x f'(x)) dx \\ &= x f(x) = \end{aligned}$$

9. **Ans (A)**

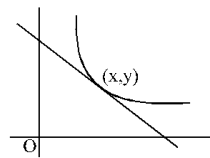
$$\begin{aligned} & (P_n)^{\frac{-1}{n^2}} \\ &= \left[\left(1 + \frac{1}{n^2}\right)^{\frac{1}{n}} \cdot \left(1 + \frac{2^2}{n^2}\right)^{\frac{2}{n}} \dots \left(1 + \frac{n^2}{n^2}\right)^{\frac{n}{n}} \right]^{\frac{-1}{n^2}} \\ &\therefore \lim_{n \rightarrow \infty} (P(n))^{\frac{-1}{n^2}} \\ &= e^{\lim_{n \rightarrow \infty} \frac{-1}{n} \sum_{r=1}^n \frac{r}{n} \log \left(1 + \frac{r^2}{n^2}\right)} \\ &= e^{-\int_0^1 x \ln(1+x^2) dx} \\ &= \frac{\sqrt{e}}{2} \end{aligned}$$

10. **Ans (D)**

$$\begin{aligned} & \text{Put } \frac{-\theta}{2} = x, \text{ we get} \\ I &= -2 \int_{-\pi/4}^{-\pi/3} \frac{e^x \sqrt{1 + \sin 2x}}{1 + \cos 2x} dx \\ &= 2 \int_{-\pi/3}^{-\pi/4} \frac{e^x |\cos x + \sin x|}{2 \cos^2 x} dx \\ &= - \int_{-\pi/3}^{-\pi/4} e^x (\sec x + \sec x \cdot \tan x) dx \\ &= -(e^x \cdot \sec x)_{-\pi/3}^{-\pi/4} \\ &= - \left(e^{-\pi/4} \sqrt{2} - e^{-\pi/3} \cdot 2 \right) \\ \therefore I &= \left(2 \cdot e^{-\pi/3} - \sqrt{2} \cdot e^{-\pi/4} \right) \end{aligned}$$

11. **Ans (C)**

$$\begin{aligned} Y - y &= m(X - x) \\ \text{for X-intercept } Y &= 0 \\ X &= x - \frac{y}{m} \\ \text{hence } x - \frac{y}{m} &= y \end{aligned}$$



$$\begin{aligned} \text{or } \frac{dy}{dx} &= \frac{y}{x - y} \\ x dy - y dy &= y dx \\ \Rightarrow -y dy &= y dx - x dy \\ \frac{-y dy}{y^2} &= \frac{y dx - x dy}{y^2} \\ \Rightarrow \frac{-dy}{y} &= d\left(\frac{x}{y}\right) \\ \text{Hence } -\ln y &= \frac{x}{y} + C \\ \text{At } x = 1, y = 1 &\Rightarrow C = 1 \\ \text{Hence } y &= e \cdot e^{-x/y} \\ e^{-x/y} &= \frac{e}{y} \end{aligned}$$

$$y e^{x/y} = e$$

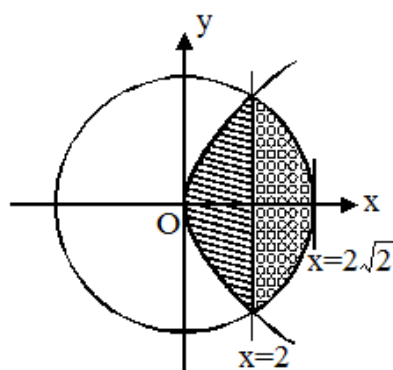
12. **Ans (D)**

$$\begin{aligned} \text{Let } y &= mx - 2am - am^3, (a = 1) \text{ touches } x^2 - y^2 = 1 \\ \Rightarrow (-2m - m^3)^2 &= (m^2 - 1) \\ \Rightarrow m^6 + 4m^4 + 3m^2 + 1 &= 0 \Rightarrow m \in \phi \end{aligned}$$

13. **Ans (B)**

$$\begin{aligned} \text{Let } I &= \int_{-\pi/2}^{\pi/2} \frac{\cos^2 x dx}{1 + e^x} \dots (i) \\ \text{Applying king property} \\ I &= \int_{-\pi/2}^{\pi/2} \frac{\cos^2 x dx}{1 + e^{-x}} \dots (ii) \\ \text{Adding (i) \& (ii)} \\ 2I &= \int_{-\pi/2}^{\pi/2} \cos^2 x dx \\ \Rightarrow 2I &= 2 \int_0^{\pi/2} \cos^2 x dx \Rightarrow I = \frac{\pi}{4} \end{aligned}$$

14. Ans (C)



$$\begin{aligned} \text{required area} &= 2 \int_0^2 \sqrt{2x} \, dx + 2 \int_2^{2\sqrt{2}} \sqrt{8-x^2} \, dx \\ &= \left(2\pi + \frac{4}{3}\right) \end{aligned}$$

15. Ans (A)

$$\frac{f(x)}{1+x^2} = 1 + \int_0^x \frac{f^2(t)}{1+t^2} \, dt \quad f(0) = 1$$

differentiate both sides w.r.t. x

$$\begin{aligned} \frac{(1+x^2)f'(x) - f(x)(2x)}{(1+x^2)^2} &= \frac{f^2(x)}{(1+x^2)} \\ \Rightarrow \frac{dy}{dx} - \left(\frac{2x}{1+x^2}\right)y &= y^2 \Rightarrow \frac{1}{y^2} \frac{dy}{dx} - \left(\frac{2x}{1+x^2}\right) \frac{1}{y} \\ &= 1 \end{aligned}$$

$$\text{Let } \frac{-1}{y} = t \Rightarrow \frac{dt}{dx} + \left(\frac{2x}{1+x^2}\right)t = 1$$

solving the above LDE,

$$f(x) = \frac{-3(1+x^2)}{x^3+3x-3}$$

$$f(-2) = \frac{-3(5)}{-8-6-3} = \frac{15}{17}$$

16. Ans (D)

$$\begin{aligned} \int \frac{dx}{(x^2+1)(x^2+9)} &= \frac{1}{8} \left[\tan^{-1}x - \frac{1}{3} \tan^{-1}\left(\frac{x}{3}\right) \right] - f(x) \\ f(\sqrt{3}) &= \frac{1}{8} \left(\frac{\pi}{3} - \frac{1}{3} \frac{\pi}{6} \right) = \frac{5\pi}{144} \end{aligned}$$

17. Ans (C)

$$\begin{aligned} \frac{ydx - xdy}{x^2 + y^2} &= \frac{dz}{z} \\ \Rightarrow \frac{\left(\frac{ydx - xdy}{y^2}\right)}{1 + \left(\frac{x}{y}\right)^2} &= \frac{dz}{z} \\ \Rightarrow \frac{d\left(\frac{x}{y}\right)}{1 + \left(\frac{x}{y}\right)^2} &= \frac{dz}{z} \\ \Rightarrow \tan^{-1}\left(\frac{x}{y}\right) &= \ln z + \ln k_1 \\ \Rightarrow z &= K e^{\tan^{-1}\left(\frac{x}{y}\right)} \end{aligned}$$

18. Ans (C)

Let equation of tangent is $y = mx + \frac{16}{m}$

It passes through (h, k)

$$\text{then } k = mh + \frac{16}{m}$$

$$\Rightarrow m^2h - mk + 16 = 0$$

It will have roots $m_1, 8m_1$

$$m_1 + 8m_1 = \frac{k}{h} \text{ \& } m_1 \cdot 8m_1 = \frac{16}{h}, \text{ eliminating } m_1,$$

we get

$$\Rightarrow \frac{k^2}{2h} = 81$$

19. Ans (D)

$$\begin{aligned} I &= \int_0^{\frac{\pi}{2}} \frac{\sin 3x - \sin x}{\sin x + \cos x} \, dx + \int_0^{\frac{\pi}{2}} \frac{\sin x}{\sin x + \cos x} \, dx \\ I &= I_1 + I_2 \end{aligned}$$

$$\begin{aligned} I_1 &= \int_0^{\frac{\pi}{2}} \frac{2(\cos 2x) \sin x}{\sin x + \cos x} \, dx = 2 \int_0^{\frac{\pi}{2}} \sin x (\cos x - \sin x) \, dx \\ &= \int_0^{\frac{\pi}{2}} \sin 2x \, dx - 2 \int_0^{\frac{\pi}{2}} \sin^2 x \, dx \end{aligned}$$

$$\therefore I_1 = \left[\frac{-1}{2} \cos 2x \right]_0^{\frac{\pi}{2}} - 2 \cdot \frac{\pi}{4} = 1 - \frac{\pi}{2}$$

$$\text{Now king and add in } I_2 \Rightarrow 2I_2 = \frac{\pi}{2} \Rightarrow I_2 = \frac{\pi}{4}$$

$$\therefore I = I_1 + I_2 = 1 - \frac{\pi}{2} + \frac{\pi}{4}$$

20. Ans (D)

$$2 \left(x - \frac{y}{\frac{dy}{dx}} \right) = \frac{1}{x}$$

$$x - \frac{1}{2x} = \frac{y}{\frac{dy}{dx}} \Rightarrow \frac{(2x^2 - 1)}{2x} = \frac{y}{\frac{dy}{dx}}$$

$$\int \frac{dy}{y} = \int \frac{2x}{(2x^2 - 1)} dx \text{ so } y = k(2x^2 - 1)^{1/2}$$

Here $k = 1$, thus $y^2 = 2x^2 - 1$

$$2x^2 - y^2 = 1.$$

SECTION-II

1. Ans (2.00)

$$\cot^{-1}(1 - x + x^2) = \tan^{-1} \frac{1}{1 - x + x^2}$$

$$= \tan^{-1} \frac{x - (x - 1)}{1 + x(x - 1)}$$

$$= \tan^{-1} x - \tan^{-1}(x - 1).$$

$$\text{Also, } \int_0^1 \tan^{-1}(x - 1) dx$$

$$= - \int_0^1 \tan^{-1} x dx$$

$$\text{Therefore, } \int_0^1 \cot^{-1}(1 - x + x^2) dx$$

$$= \int_0^1 (\tan^{-1} x - \tan^{-1}(x - 1)) dx$$

$$= 2 \int_0^1 \tan^{-1} x dx,$$

$$\text{Hence } \frac{\int_0^1 \cot^{-1}(1 - x + x^2) dx}{\int_0^1 \tan^{-1} x dx}$$

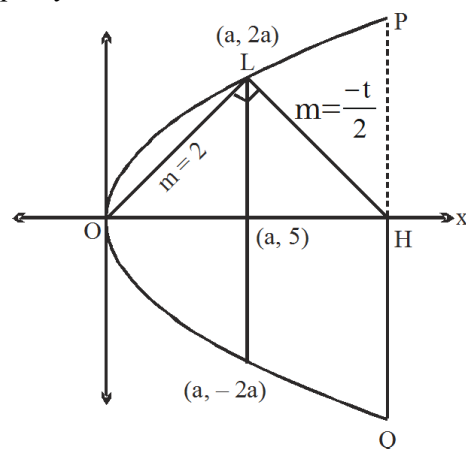
$$= 2$$

2. Ans (4.00)

Equation of LH [T/S]

$$y - 2a = \frac{-1}{2}(x - a)$$

put $y = 0$



Slope of OL = 2

Slope of LH = $-\frac{1}{2}$

$$x = 5a$$

$$\therefore y^2 = 4ax = (4a)(5a) = 20a^2$$

$$y = 2\sqrt{5}a$$

$$2y = 4\sqrt{5}a$$

3. Ans (1.00)

A tangent to $\frac{x^2}{16} - \frac{y^2}{9} = 1$ is

$$y = m_1 x \pm \sqrt{16m_1^2 - 9} \quad \text{.....(i)}$$

The other hyperbola is $\frac{x^2}{-9} - \frac{y^2}{-16} = 1$

then tangent to it is

$$y = m_2 x \pm \sqrt{-9m_2^2 - (-16)} \quad \text{.....(ii)}$$

(i) & (ii) are same.

$$\Rightarrow m_1 = m_2 \text{ \& } 16m_1^2 - 9 = -9m_2^2 + 16$$

$$\Rightarrow 25m_1^2 = 25 \Rightarrow m_1^2 = 1 \Rightarrow m_1 = \pm 1$$

$$\therefore m = 1$$

4. Ans (3.00)

$$\text{Let } I = \int 2 \cdot \sin 2\theta \cdot \cos^2 \theta \cdot e^{\sin \theta} d\theta$$

$$= \int 4 \sin \theta \cdot \cos^3 \theta \cdot e^{\sin \theta} \cdot d\theta$$

Put $\sin \theta = t$, so

$$I = \int 4t(1-t^2)e^t dt$$

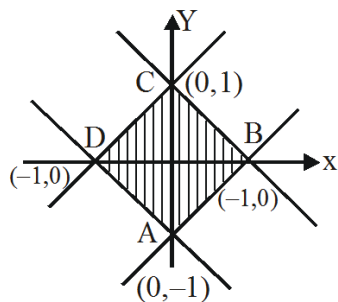
$$= 4 \int \underbrace{t}_{I_1} \cdot \underbrace{e^t}_{II} - 4 \int t^3 \cdot e^t dt$$

$$\therefore I = 4I_1 - 4I_2 \quad \dots(1)$$

$$\text{We get, } I = -4 \sin^3 \theta - 12 \cos^2 \theta - 20 \sin \theta + 32;$$

$$\therefore A = -4, B = -12, C = -20, D = 0, E = 32$$

5. Ans (2.00)



Area of shaded region

$$ABCD = 2 \cdot \frac{1}{2} \cdot 2 \cdot 1$$

$$= 2$$

6. Ans (16.00)

The equation of normal at

$$(\sqrt{6} \sec \theta, \sqrt{2} \tan \theta) \text{ is } \frac{\sqrt{6}x}{\sec \theta} + \frac{\sqrt{2}y}{\tan \theta} = 8$$

$$\text{Slope} = -1 \Rightarrow \frac{-\sqrt{6}}{\sec \theta} \times \frac{\tan \theta}{\sqrt{2}} = -1$$

$$\sin \theta = \frac{1}{\sqrt{3}}$$

$$N : x + y = 4$$

$$\text{Now, } y = -x + 4 \text{ is tangent to } \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\text{Hence, } (a^2 + b^2) = 16$$

7. Ans (7.00)

$$\int_{\ln 2}^{\ln 3} f(x) dx + \int_8^{27} g(y) dy = 27 \ln 3 - 8 \ln 2$$

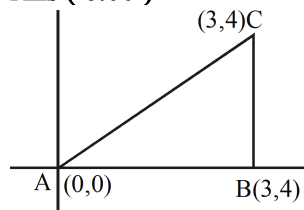
$$\text{and } \int_{\ln 2}^{\ln 3} f(x) dx = 12 - 12 \ln 3 + 12 \ln 2$$

$$\therefore \int_8^{27} g(y) dy = 39 \ln 3 - 20 \ln 2 - 12$$

$$\text{Hence, } a = 39; b = 20; c = 12$$

$$\therefore a - (b + c) = 39 - 32 = 7$$

8. Ans (6.00)



$$AC = 5 = 2a$$

$$\Rightarrow 2ae = 5 \quad \dots(1)$$

$$\text{Also } AB + BC = 2a$$

$$\Rightarrow 2a = 7$$

$$\Rightarrow a = \frac{7}{2}$$

$$\therefore e = \frac{5}{7} \Rightarrow 1 - \frac{b^2}{a^2} = \frac{25}{49} \Rightarrow b = \sqrt{6}$$

$$\text{Area} = \pi ab = \pi \times \frac{7}{2} \times \sqrt{6}$$

$$\therefore P = 6$$

9. Ans (1.00)

$$\text{We have } \lim_{x \rightarrow 0} 2 \left(\frac{\sqrt[3]{1 + \sin^2 x} - \sqrt[4]{1 - 2 \tan x}}{\sin x + \tan^2 x} \right)$$

$$= 1$$

10. Ans (500.00)

$$f'(x) = f'(5 - x)$$

$$\Rightarrow f(x) + f(5 - x) = \text{constant} = k$$

$$\Rightarrow -10 + 50 = k \Rightarrow k = 40$$

$$\text{Now } I = 5 \int_0^5 f(x) dx$$

$$I = 5 \int_0^5 f(5 - x) dx$$

$$2I = 5 \int_0^5 [f(x) + f(5 - x)] dx$$

$$2I = 5.40.5$$

$$I = 5.20.5$$

$$I = 500$$